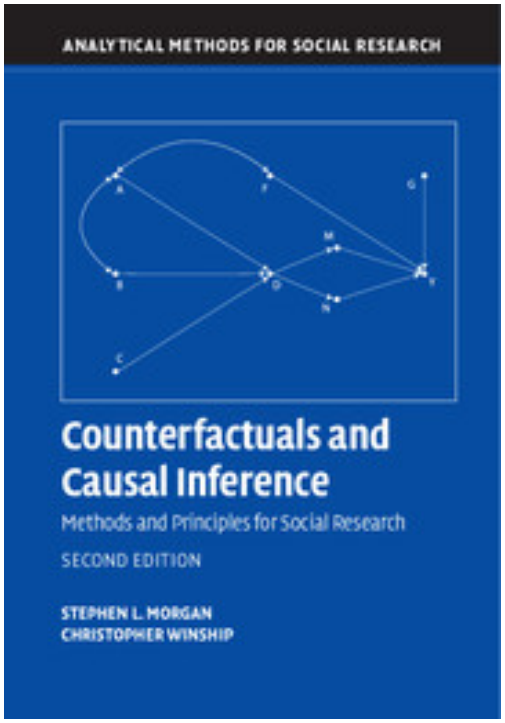
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 2

Outline

1. Defining the Causal States
2. Potential Outcomes and Individual-Level Treatment Effects
3. Treatment Groups and Observed Outcomes
4. The Average Treatment Effect
5. The Stable Unit Treatment Value Assumption
6. Treatment Assignment and Observational Studies
7. Average Causal Effects and Naive Estimation
8. Over-Time Potential Outcomes and Causal Effects
9. The Potential Outcome Model for Many-Valued Treatments
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Fine articulation

Worker training example

"entered a training program" (the treatment state)
"did not enter a training program" (the control state)

If treatment-site heterogeneity exists, then stratified analyses may be necessary, perhaps by regions of the country, size of the program or school, or whatever other dimension suggests that variability of the causal states deserves explicit modeling.

Nominal states from constitutive features

Each state of each treatment should be regarded as a nominal state with constitutive features (e.g., entities, activities, and relations) that are jointly capable of producing the outcome of interest.

Catholic school example with learning as the outcome

Nominal states: "enrolled in a Catholic school" and "enrolled in a public school"
Each type of school has: teachers, classrooms, curricula, administrators, normative environments, affiliated institutions, and networks of peers and parents.

Local and reasonable

- The "what would have been" nature of the conditionals that define the causal states of interest be carefully considered.
- When a facile *ceteris paribus* assumption is invoked to relieve the analyst from having to discuss other contrasts that are nearly certain to occur at the same time, the posited causal states may be open to the charge that they are too improbable or ill-defined to justify the pursuit of a causal analysis based on them.

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Potential outcomes and treatment effects

The individual-level treatment effect

$$\delta_i = y_i^1 - y_i^0. \tag{2.1}$$

- There is a wide variety of notation in the potential outcome and counterfactuals literature. Equation (2.1) and its elements are often written as one of the following alternatives,

$$\begin{aligned} \Delta_i &= Y_{1i} - Y_{0i}, \\ \delta_i &= Y_i^t - Y_i^c, \\ \tau_i &= y_i(1) - y_i(0), \end{aligned}$$

  and variants thereof.
- Different ways of defining individual-level causal effects exist as well, such as y_i^1/y_i^0.

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Observed outcomes

- The random variable D takes on values of $d_i = 1$ for each individual i who is an observed member of the treatment group and $d_i = 0$ for each individual i who is an observed member of the control group.
- We observe values for a variable Y as $y_i = y_i^1$ for individuals with $d_i = 1$ and as $y_i = y_i^0$ for individuals with $d_i = 0$.
- The observable outcome variable Y is therefore defined as

$$\begin{aligned} Y &= Y^1 \text{ if } D = 1, \\ Y &= Y^0 \text{ if } D = 0. \end{aligned}$$

This paired definition is often written compactly as

$$Y = DY^1 + (1 - D)Y^0. \tag{2.2}$$

Table 2.1 The Fundamental Problem of Causal Inference

Group	Y^1	Y^0
Treatment group ($D = 1$)	Observable as Y	Counterfactual
Control group ($D = 0$)	Counterfactual	Observable as Y

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Average treatment effects

The average treatment effect (ATE)

$$\begin{aligned} E[\delta] &= E[Y^1 - Y^0] \\ &= E[Y^1] - E[Y^0]. \end{aligned} \tag{2.3}$$

An alternative group-level causal effect is the causal risk ratio

$$\frac{Pr[Y^1 = 1]}{Pr[Y^0 = 1]}. \tag{2.4}$$

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Table 2.2 A Hypothetical Example in Which SUTVA Is Violated

Treatment assignment patterns			Potential outcomes	
$\begin{bmatrix} d_1 = 1 \\ d_2 = 0 \\ d_3 = 0 \end{bmatrix}$ or	$\begin{bmatrix} d_1 = 0 \\ d_2 = 1 \\ d_3 = 0 \end{bmatrix}$ or	$\begin{bmatrix} d_1 = 0 \\ d_2 = 0 \\ d_3 = 1 \end{bmatrix}$	$y_1^1 = 3$ $y_2^1 = 3$ $y_3^1 = 3$	$y_1^0 = 1$ $y_2^0 = 1$ $y_3^0 = 1$
$\begin{bmatrix} d_1 = 1 \\ d_2 = 1 \\ d_3 = 0 \end{bmatrix}$ or	$\begin{bmatrix} d_1 = 0 \\ d_2 = 1 \\ d_3 = 1 \end{bmatrix}$ or	$\begin{bmatrix} d_1 = 1 \\ d_2 = 0 \\ d_3 = 1 \end{bmatrix}$	$y_1^1 = 2$ $y_2^1 = 2$ $y_3^1 = 2$	$y_1^0 = 1$ $y_2^0 = 1$ $y_3^0 = 1$

Individual-level treatment effects without SUTVA

- Suppose that $\mathbf{d}$ is an $N \times 1$ vector of treatment indicator variables for N individuals
- Define the potential outcomes of each individual as functions across all potential configurations of the elements of vector $\mathbf{d}$.
- The outcome for individual i under the treatment is $y_i^1(\mathbf{d})$, and the outcome for individual i under the control is $y_i^0(\mathbf{d})$.
- The treatment effect for each individual i is then

$$\delta_i(\mathbf{d}) = y_i^1(\mathbf{d}) - y_i^0(\mathbf{d}). \tag{2.5}$$

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Independence of potential outcomes

- For randomized experiments, the treatment indicator variable D is forced by design to be independent of the potential outcome variables Y^1 and Y^0.
- Treatment status is independent of the potential outcomes, and the treatment assignment mechanism is said to be ignorable. This independence assumption is usually written as

$$(Y^0, Y^1) \perp\!\!\!\perp D, \tag{2.6}$$

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Conditional average treatment effects

- The ATT is the average treatment effect for those who typically take the treatment:

$$\begin{aligned} E[\delta|D=1] &= E[Y^1 - Y^0|D=1] \\ &= E[Y^1|D=1] - E[Y^0|D=1], \end{aligned} \tag{2.7}$$

- The ATC is the average treatment effect for those who typically take the control:

$$\begin{aligned} E[\delta|D=0] &= E[Y^1 - Y^0|D=0] \\ &= E[Y^1|D=0] - E[Y^0|D=0], \end{aligned} \tag{2.8}$$

Naive estimation of average treatment effects

The naive estimator

$$\hat{\delta}_{\text{NAIVE}} \equiv E_N[y_i|d_i = 1] - E_N[y_i|d_i = 0], \tag{2.9}$$

- The naive estimator converges to $E[Y|D = 1] - E[Y|D = 0]$.
- In observational studies, the naive estimator rarely yields consistent or unbiased estimates of the ATE.

Inconsistency and bias of the naive estimator

- To begin to see the predicament, first decompose the ATE in Equation (2.3) as

$$E[\delta] = \{\pi E[Y^1|D=1] + (1-\pi)E[Y^1|D=0]\} \quad (2.10)$$
$$- \{\pi E[Y^0|D=1] + (1-\pi)E[Y^0|D=0]\}.$$

- Equation (2.10) reveals that the ATE is a function of five unknowns: the proportion of the population that is assigned to the treatment along with four conditional expectations of the potential outcomes.

- With observational data from a random sample of the population, we can compute estimates that are consistent and unbiased for only three of the five unknowns on the right-hand side of Equation (2.10).
- Consider π first, which we have defined as equal to $E[D]$, and which is the fixed proportion of the population assigned to the treatment.
- The sample-mean estimator, $E_N[d_i]$, is consistent for π, which we write as

$$E_N[d_i] \xrightarrow{p} \pi. \tag{2.11}$$

- Equation (2.11) represents the claim that, as the sample size N increases to infinity, the sample mean of the values for d_i converges to the true value of π, which we assume is a fixed population parameter equal exactly to $E[D]$.

- We can offer similar claims for consistent estimators of two other unknowns in Equation (2.10):

$$E_N[y_i|d_i = 1] \xrightarrow{p} E[Y^1|D = 1], \tag{2.12}$$

$$E_N[y_i|d_i = 0] \xrightarrow{p} E[Y^0|D = 0], \tag{2.13}$$

  which indicate that the sample mean of the observed outcome in the treatment group converges to the true average outcome under the treatment state for those in the treatment group (and analogously for the control group and control state).
- But we still have two unknowns: the counterfactual conditional expectations of the potential outcomes: $E[Y^1|D = 0]$ and $E[Y^0|D = 1]$.

An example of two sources of bias

Consider this rearrangement of the decomposition in Equation (2.10):

$$\begin{aligned} E[Y^1|D=1] - E[Y^0|D=0] = E[\delta] & \\ + \ \{E[Y^0|D=1] - E[Y^0|D=0]\} & \\ + \ (1-\pi)\{E[\delta|D=1] - E[\delta|D=0]\}. & \end{aligned} \tag{2.14}$$

- The first source, $\{E[Y^0|D=1] - E[Y^0|D=0]\}$, is a *baseline bias* equal to the difference in the expected outcome in the absence of the treatment between those in the treatment group and those in the control group.
- The second source, $(1-\pi)\{E[\delta|D=1] - E[\delta|D=0]\}$, is a *differential treatment effect bias* equal to the expected difference in the treatment effect between those in the treatment group and those in the control group.

Table 2.3 An Example of Inconsistency and Bias of the Naive Estimator When the ATE Is the Causal Effect of Interest

Group	$E[Y^1\|D]$	$E[Y^0\|D]$
Treatment group ($D = 1$)	10	6
Control group ($D = 0$)	8	5

If the proportion of the population that has a bachelor's degree is .3, then the ATE is 3.3, which is equal to
$.3(10 - 6) + (1 - .3)(8 - 5)$.

- The naive estimator, as defined in Equation (2.9), would be equal to 5 for an infinite sample (or equal to 5, on average, across repeated samples). Thus, the naive estimator is inconsistent and upwardly biased for the ATE (i.e., yielding 5 rather than 3.3), the ATT (i.e., yielding 5 rather than 4), and the ATC (i.e., yielding 5 rather than 3).
- Equation (2.14) gives the components of the total expected bias of 1.7 for the naive estimator as an estimate of the ATE. The term $\{E[Y^0|D=1] - E[Y^0|D=0]\}$, which we labeled the expected baseline bias, is $6 - 5 = 1$. The term $(1-\pi)\{E[\delta|D=1] - E[\delta|D=0]\}$, which is the expected differential treatment effect bias, is $(1-.3)(4-3) = .7$.

Estimating causal effects under maintained assumptions about potential outcomes

- We can identify the ATE if we can accept the following assumptions about potential outcomes for subsets of the population defined by treatment status

$$\text{Assumption 1:} \quad E[Y^1|D=1] = E[Y^1|D=0] \tag{2.15}$$

$$\text{Assumption 2:} \quad E[Y^0|D=1] = E[Y^0|D=0] \tag{2.16}$$

- Or we can estimate the ATT or the ATC by assuming one but not the other.

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A single unit over time

- Consider the analysis of a single unit, observed during time intervals indexed by a discrete counter t that increases from 1 to T.
- The outcome variable is Y_t, which has observed values $\{y_1, y_2, y_3, ..., y_T\}$.
- Suppose that we have a two-state causal variable, D_t, that is equal to 1 if the treatment is in place during a time period t and is equal to 0 otherwise.

For a single unit, the causal effect of the treatment is

$$\delta_t = Y_t^1 - Y_t^0 \tag{2.17}$$

1. Before the treatment is introduced (for $t < t^*$):

$$\begin{aligned} D_t &= 0 \\ Y_t &= Y_t^0 \end{aligned}$$

2. While the treatment is in place (from t^* through $t^* + k$):

$$\begin{aligned} D_t &= 1 \\ Y_t &= Y_t^1 \end{aligned}$$

Y_t^0 exists but is counterfactual

3. After the treatment ends (for time periods $t > (t^* + k)$) :

$$\begin{aligned} D_t &= 0 \\ Y_t &= Y_t^1 \end{aligned}$$

Y_t^0 exists but is counterfactual.

The year of the fire horse

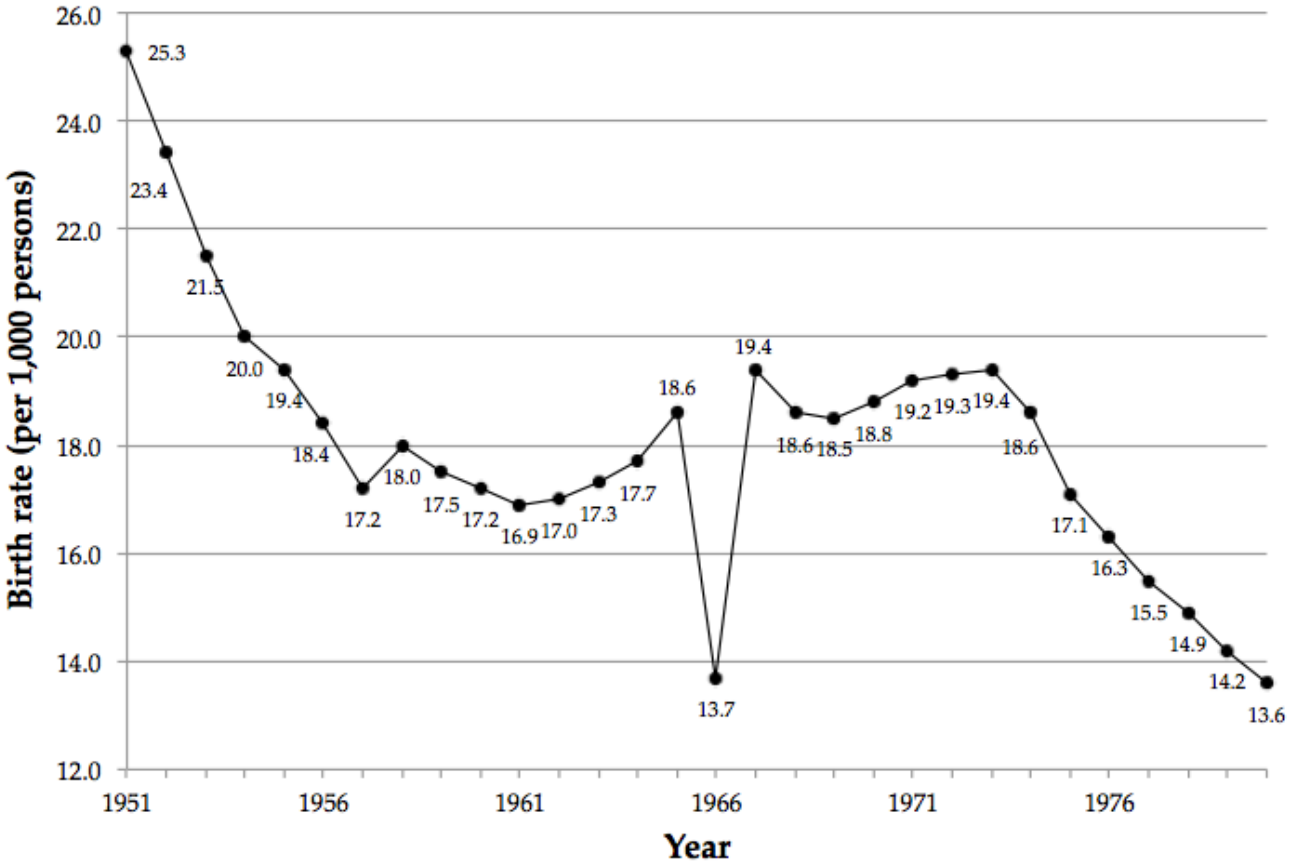

Figure 2.1 Crude birth rates in Japan, 1951-1980.
Source: Table 2-24, Live Births by Sex and Sex Ratio of Live Births, Bureau of Statistics of Japan. Accessed at http://www.stat.go.jp/data/chouki/zuhyou/02-24.xls on March 11, 2013.

- The states are "fire-horse year in the zodiac calendar" and "not," and they are aligned, not with observed treatment and control groups, but with the observed years, $D_{1966} = 1$ versus $D_{t \neq 1966} = 0$.
- In the year 1966, the treatment generates two corresponding potential outcome variables, Y^1_{1966} and Y^0_{1966} , which define the effect of interest for the birth rate in Japan in 1966:

$$\delta_{1966} = y^1_{1966} - y^0_{1966}.$$

- We can estimate this effect as

$$\begin{aligned} \hat{\delta}_{1966} &= y_{1966} - \hat{y}^0_{1966} \\ &= 13.7 - \hat{y}^0_{1966}. \end{aligned}$$

- The estimate $\hat{\delta}_{1966}$ is the observed birth rate in 1966, 13.7, minus an estimate of the counterfactual birth rate, $\hat{y}^0_{1966}$, that would have been observed if 1966 had not been the year of the fire horse.

- What is a reasonable estimated value for the true counterfactual outcome, y^0_{1966}?
- One reasonable choice: the average of the two values for 1963 and 1969,

$$(17.3 + 18.5)/2 = 17.9$$

- The result of such a choice is an estimated causal effect,

$$\hat{\delta}_{1966} = 13.7 - 17.9 = -4.2,$$

that implies a decline in the crude birth rate of 23 percent.

- For this estimate, we are assuming positive fire-horse-year effects in 1964, 1965, 1967, and 1968, which were non–fire-horse years.
- As a result, in order to estimate the overall effect of the year of the fire horse on the population structure of Japan, we would need to model and then appropriately combine five year-specific effects, δ_{1964} through δ_{1968}, based on additional corresponding treatment states for the specific years.

Many units over time

- Unlike for a single unit, we will not utilize notation for a focal time interval, t^*, in which the treatment is introduced.
- Instead, we want to preserve the possibility that the treatment is introduced at different times for different individuals.
- We will also now distinguish between two different treatment indicator variables.
  - D_{it} is a time-varying *treatment exposure indicator* that indicates whether individual i receives the treatment in time period t.
  - D_i^* is a time-constant *treatment group indicator* that indicates whether individual i ever receives the treatment at any point in time during the observation window of the study (i.e., in any time period from $t = 1$ to T).

- For members of the control group, $Y_{it} = Y_{it}^0$ for all time periods t.
- For members of the treatment group, $Y_{it} = Y_{it}^0$ before treatment exposure, and $Y_{it} = Y_{it}^1$ after treatment exposure begins.
- Altogether, Y_{it} is defined with reference to D_{it}, such that

$$Y_{it} = D_{it}Y_{it}^1 + (1 - D_{it})Y_{it}^0 \tag{2.18}$$

The distinction between D_{it} and D_i^* reveals the potential value of panel data.

- For a cross-sectional study in which all observation occurs in a single time period, $D_{it} = D_i^*$.
- For a panel dataset over multiple time periods, a researcher may be able to consider how treatment exposure (D_{it}) can be separated from treatment group membership (D_i^*) and then exploit this difference to estimate the causal effect.

- It may be possible to make reasonable predictions about how Y_{it}^0 would have evolved over time for individuals in the treatment group during the posttreatment time period.
- The crux of the matter, of course, is how to use observed values of $Y_{it} = Y_{it}^0$ in time period t among those in the control group ($D_i^* = 0$) to make reasonable predictions about posttreatment counterfactual values of Y_{it}^0 for those in the treatment group ($D_i^* = 1$).

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- Consider the more general setup:
  1. a set of J treatment states,
  2. a corresponding set of J causal exposure dummy variables, $\{Dj\}_{j=1}^{J}$, and
  3. a corresponding set of J potential outcome random variables, $\{Y^{Dj}\}_{j=1}^{J}$.
- Each individual receives only one treatment, which we denote Dj'.
- The observed outcome variable for individual i, y_i, is then equal to $y_i^{Dj'}$. For the other $J-1$ treatments, the potential outcomes of individual i exist in theory as $J-1$ other potential outcomes y_i^{Dj} for $j \neq j'$, but they are counterfactual.

Table 2.4 The Fundamental Problem of Causal Inference for Many-Valued Treatments

Group	Y^{D1}	Y^{D2}	$\cdots$	Y^{DJ}
Takes $D1$	Observable as Y	Counterfactual	$\cdots$	Counterfactual
Takes $D2$	Counterfactual	Observable as Y	$\cdots$	Counterfactual
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
Takes DJ	Counterfactual	Counterfactual	$\cdots$	Observable as Y

Table 2.5 The Observability Table for Estimating How Education Increases Earnings

Education	Y^{HS}	Y^{AA}	Y^{BA}	Y^{MA}
Obtains HS	Observable as Y	Counterfactual	Counterfactual	Counterfactual
Obtains AA	Counterfactual	Observable as Y	Counterfactual	Counterfactual
Obtains BA	Counterfactual	Counterfactual	Observable as Y	Counterfactual
Obtains MA	Counterfactual	Counterfactual	Counterfactual	Observable as Y

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Population and data generation models

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- Alternative Perspectives